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New Paleomagnetic Results From the Late Mesoproterozoic Luanshigou Formation, Shennongjia Group in South China and Their Implications for the Pre-Grenvillian Connections Between South China Blocks and Southwestern Laurentia

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Abstract The identification of the Grenvillian-age ophiolite suites in the Yangtze block in recent years suggested that the northern Yangtze subblock (NYB) and the southern Yangtze subblock (SYB) were once separated by an ocean in late Mesoproterozoic. Although some paleogeographic models advocated the Pre-Grenvillian connections between the south China blocks and Laurentia, none of them has been paleomagnetically tested. Here we report the new paleomagnetic results obtained from the ~1,270 Ma purplish-red muddy dolomite beds of the Luanshigou Formation, Shennongjia Group, NYB, providing new constraints for reconstructing the paleogeographic positions of the south China blocks in late Mesoproterozoic. A total of 447 samples underwent stepwise thermal demagnetization. Two components were identified. The low-temperature component is interpreted as the recent viscous remnant magnetization. The high-temperature component was obtained from 64 samples below 580°C and from 177 samples below 690°C, directed northeast-up or southwest-down, antipodally, positioning the paleomagnetic pole at 18.5°S, 74.4°E (dm/dp = 2.5/1.6°). Rock magnetic results demonstrate that the magnetic carriers in purplish-red dolomite and pale-pink dolomite are predominated by hematite and magnetite, respectively. The data quality is supported by an inverse baked contact test, a B-class reversal test, and the paleomagnetic pole is distinct from any younger poles of the region. Based on the paleomagnetic results, aided by geological evidence, we propose a reconstruction in which the NYB was juxtaposed to southwestern Laurentia in the late Mesoproterozoic and suggest that the late Mesoproterozoic Miaowan-Shimian ophiolite zone in the Yangtze block was likely an extension of the Grenville belt of Laurentia.

Plain Language Summary The south China block was traditionally subdivided into the Yangtze block to the northwest and the Cathaysia block to the southeast, which were separated by an early Neoproterozoic ocean. Recently published geological data further suggest that the Yangtze block contains two subblocks, the NYB and the SYB, which were separated by a late Mesoproterozoic ocean. Several paleogeographic models proposed the Mesoproterozoic relationships between the south China blocks and Laurentia, but none has been paleomagnetically tested. Here we report a new paleomagnetic pole obtained from the ~1,270 Ma purplish-red dolomite beds of the Luanshigou Formation, Shennongjia Group in the NYB. The data quality is assured by the positive paleomagnetic field tests. Our results combing with the geological evidence suggest that the NYB was located in a low-latitudinal position and connected to southwestern Laurentia in the late Mesoproterozoic. The late Mesoproterozoic ophiolite zone in the southern margin of the NYB was likely an extension of the Grenville belt.

Key Points:

- A new paleomagnetic pole (18.5°S, 74.4°E, dm/dp = 2.5/1.6°) was obtained from the late Mesoproterozoic Luanshigou Formation of the Shennongjia Group in south China
- The northern Yangtze subblock (NYB) was connected to southwestern Laurentia at the late Mesoproterozoic
- The late Mesoproterozoic Miaowan-Shimian suture zone in the southern margin of the NYB was likely an extension of the Grenvillian orogenic belt

Supporting Information:

Supporting Information may be found in the online version of this article.

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1. Introduction

The south China block was traditionally subdivided into the Yangtze block to the northwest and the Cathaysia block to the southeast, which are separated by the early Neoproterozoic Jiangnan Orogen (e.g., H. Wang et al., 2005; G. Zhao, 2015). In recent years, the identification of a Grenvillian-age ophiolite suture zone (see the Shimian-Miaowan suture in Figure 1a) within the Yangtze block indicated that there was a late Mesoproterozoic ocean once separated the northern Yangtze subblock (NYB) from the southern Yangtze subblock (SYB)

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(Figure 1a; Deng et al., 2017; X. Jiang et al., 2016; Lu et al., 2020). It is suggested that a coherent Yangtze block had been formed by ~860 Ma through the collision of the SYB and the NYB (Lu et al., 2020).

In some recent paleocontinental reconstruction models, the Yangtze block was placed close to the northwestern Laurentia in the Nuna supercontinent based on the correlation of the Mesoproterozoic rocks from southwestern SYB and northwestern Laurentia (Cawood et al., 2018, 2020; Gladkochub et al., 2021; W. Wang et al., 2016). In other models, however, the Cathaysia block has been placed close to western Laurentia in the Nuna supercontinent based on the similarities of Mesoproterozoic magmatic events (~1,460–1,430 Ma) and metamorphic records (~1,300–1,000 Ma) in the Hainan terrane and southwestern Laurentia, and the provenance analyses of the Precambrian strata in south China (e.g., Z. X. Li et al., 1995, 2008; Y. J. Xu et al., 2020; S. H. Zhang et al., 2012), although the debate that whether the Mesoproterozoic Hainan terrane was part of the Cathaysia block or the SYB still stands (Cawood et al., 2020; Y. J. Xu et al., 2020). None of these reconstruction models has been paleomagnetically tested yet.

In this paper, we report the paleomagnetic results newly obtained from the ~1,270 Ma Luanshigou Formation (Fm) of the Shennongjia Group (Gp) in the NYB. The paleomagnetic data, combined with geological evidence, permit that the NYB was connected to the southwestern Laurentia in the time around 1,270 Ma.

2. Geological Background and Sampling

The Shennongjia Gp largely outcrops in the Shennongjia area in the western Hubei Province, ~13 km northwestern of the Archean-Paleoproterozoic Kongling complex that is considered as the oldest basement core of the NYB (Figures 1a and 1b; S. Gao et al., 1999, 2011; Guo et al., 2015; Y. Qiu et al., 2000; S. Zhang et al., 2006a, 2006b). It is composed mainly of stromatolite carbonates and intercalated clastic rocks, with minor interbedded basic volcanic rocks and tuff beds presenting in the middle and the upper part of the group (Figures 1c and 2; Kuang et al., 2018; Q. Li & Leng, 1991; Liu et al., 2004).

The Shennongjia Gp is unconformably overlain by the typical Neoproterozoic successions of the Yangtze block (Z. Hu et al., 2012; Kuang et al., 2019; X. M. Zhao et al., 2019). The ~1,800–1,500 Ma Wujiatai Fm, unconformably overlaying the Kongling complex in the northern Kongling area, is considered equivalent to the lower portion of the Shennongjia Gp (X. M. Zhao et al., 2019, 2022). The provenance analyses suggest that the clastic sediments of the Shennongjia Gp were mainly sourced from the ancient basement rocks in the Yangtze block with minor probably sourced from the Cathaysia block or the Precambrian basement of Laurentia or Australia (H. Li et al., 2013; D. Xu et al., 2016; J. Wang et al., 2013). It is thus generally considered that the Shennongjia Gp is of the cover sequence developed upon the NYB.

Many efforts had been made to constrain the age of the Shennongjia Gp. A tuff bed in the middle Yemahe Fm has been dated at 1,215.8 ± 2.4 Ma (Figure 2; H. Li et al., 2013), through use of LA–ICP–MS (laser ablation–inductively coupled plasma–mass spectrometry) zircon U-Pb dating method. The volcano-sedimentary rocks and volcanic rocks near the top of the Zhengjiaya Fm have been dated at 1,103 ± 8 Ma and 1,063 ± 16 Ma, respectively (Figure 2; X. F. Qiu et al., 2011, 2015), through use of LA–ICP–MS zircon U-Pb dating method. These analyses provide the most reliable age constraints for the strata. A stromatolite sample from the upper Kuangshishan Fm has been dated at 1,285.7 ± 66.6 Ma (Figure 2; Y. Jiang et al., 2024), using the LA–ICP–MS carbonate U-Pb dating method, which provides an age constraint for the lower part of the Shennongjia Gp. Moreover, detrital zircon analyses of the sandstone of the Dayanping Fm have revealed a major age population around 1,400 Ma, with two youngest zircon grains yielded $^{207}Pb/^{206}Pb$ ages of 1,325 ± 8 Ma and 1,324 ± 27 Ma respectively, both being concordant well (Figure S1 in Supporting Information S1; H. Li et al., 2013), indicating that the strata are younger than ~1,325 Ma. A few diabase dykes intruded multifold horizons of the Shennongjia Gp (Figures 1b and 2). One of the dykes that intruded into the Shicaohe Fm was dated at 1,115 ± 9 Ma (Figure 2; H. Li et al., 2013), through use of SIMS (secondary ion mass spectrometry) baddeleyite U-Pb dating method, indicating that the host strata are older than ~1,115 Ma. Recently, a Re-Os age of 1,258.3 ± 8.5 Ma was newly obtained from the rocks near the bottom of Taizi Fm (Figure 2; B. Gao et al., 2024; Supporting Information S2), which is consistent with the geochronological frame mentioned above.

The sampled Luanshigou Fm, which is ~1,137 m thick (Figure 2), contains two members. The Lower Member is composed of pale-grayish intraclast dolomite, siliceous dolomicrite, and siliceous banded dolomite (Figure 3a).

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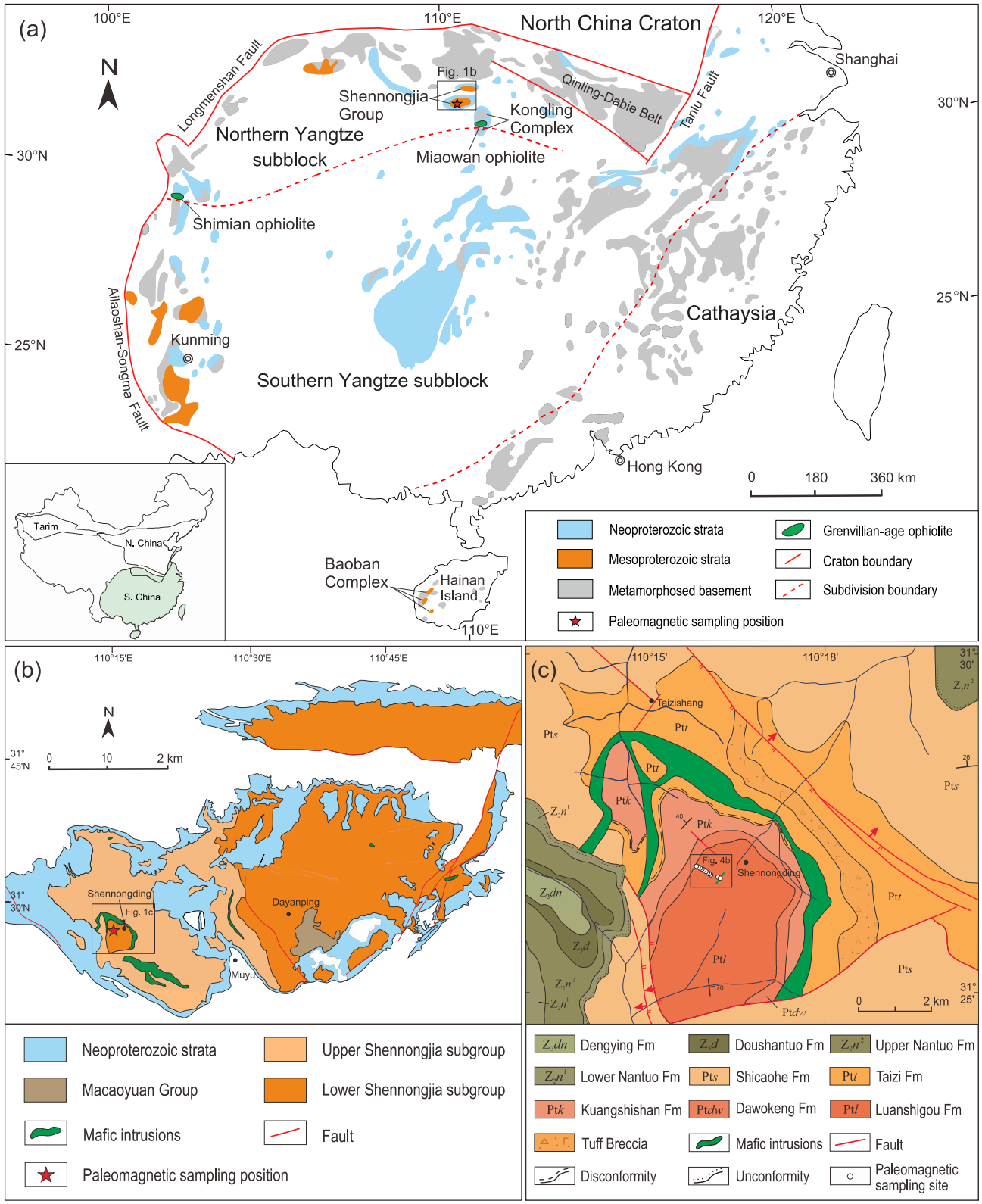


Figure 1. (a) Geological map of the south China block (SCB) showing tectonic boundaries and the distribution of the Shennongjia Group in the northern Yangtze subblock (modified from S. H. Zhang et al., 2021). Inset showing the position of the SCB. (b) Simplified geological map of the Shennongjia area showing paleomagnetic sampling location. (c) Geological map of the Shennongding area showing the location of paleomagnetic sampling sites. Fm = Formation.

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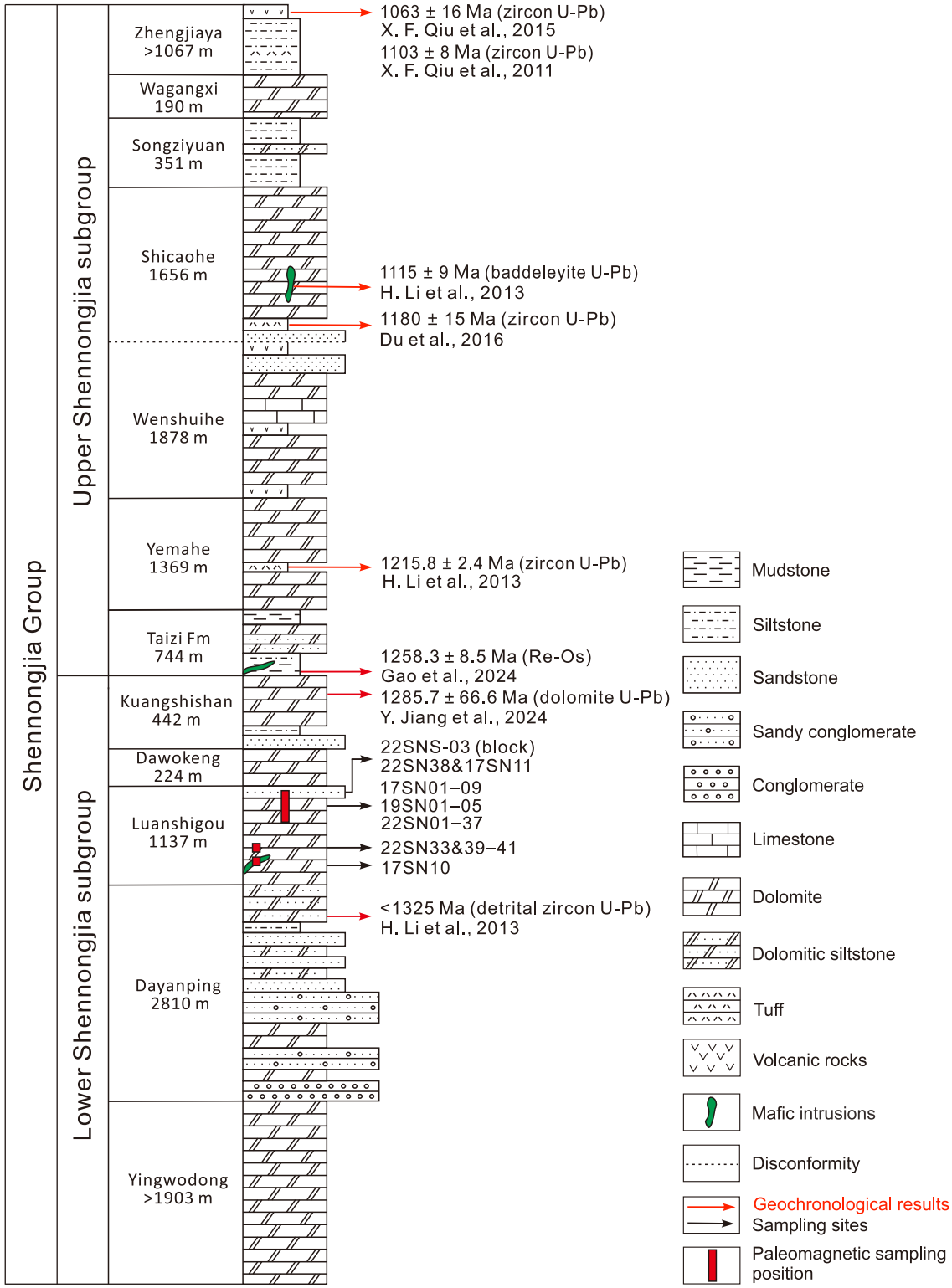


Figure 2. Stratigraphic column of the Shennongjia Group (modified from H. Li et al., 2013).

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Figure 3. (a) Stromatolite dolomite of the Luanshigou Formation. (b) The contact line of the diabase sill and sandy dolomite of the lower Luanshigou Formation in the Shennongding section. (c) Laminae in the purplish-red pelitic dolomite of the Luanshigou Formation (the marking pen is 14 cm long). (d) Representative paleomagnetic sampling outcrop of the Shennongding section.

The Upper Member mainly consists of purplish-red pelitic dolomite, siliceous banded dolomite, and sandstone near the top of the Luanshigou Fm (Figures 3c and 3d).

The paleomagnetic sampling was carried out at the Shennongding section (GPS position of section base 31.446°N, 110.389°E; section top 31.448°N, 110.263°E.) of the Shennongjia area. A total of 447 samples from 46 sites were collected from the purplish-red pelitic dolomite, sandstone, and pale-pink dolomite of the Upper Member of the Luanshigou Fm, covering a 140 m-thick stratigraphic interval (Figure 4). We collected 23 samples from the diabase sill that intruded in the Lower Member of the Luanshigou Fm, and 43 samples from the baked grayish sandy dolomite adjacent to the diabase sill within ~3 m from the contact margin (Figures 3b and 4b) for baked contact test. We collected 1 block sample from the sandstone near the top of the section for detrital zircon U-Pb dating analysis (Figure 2). All the paleomagnetic samples were cores, 2.54 cm in diameter, collected using a portable drill and oriented by using a magnetic compass and a solar compass when possible. The difference

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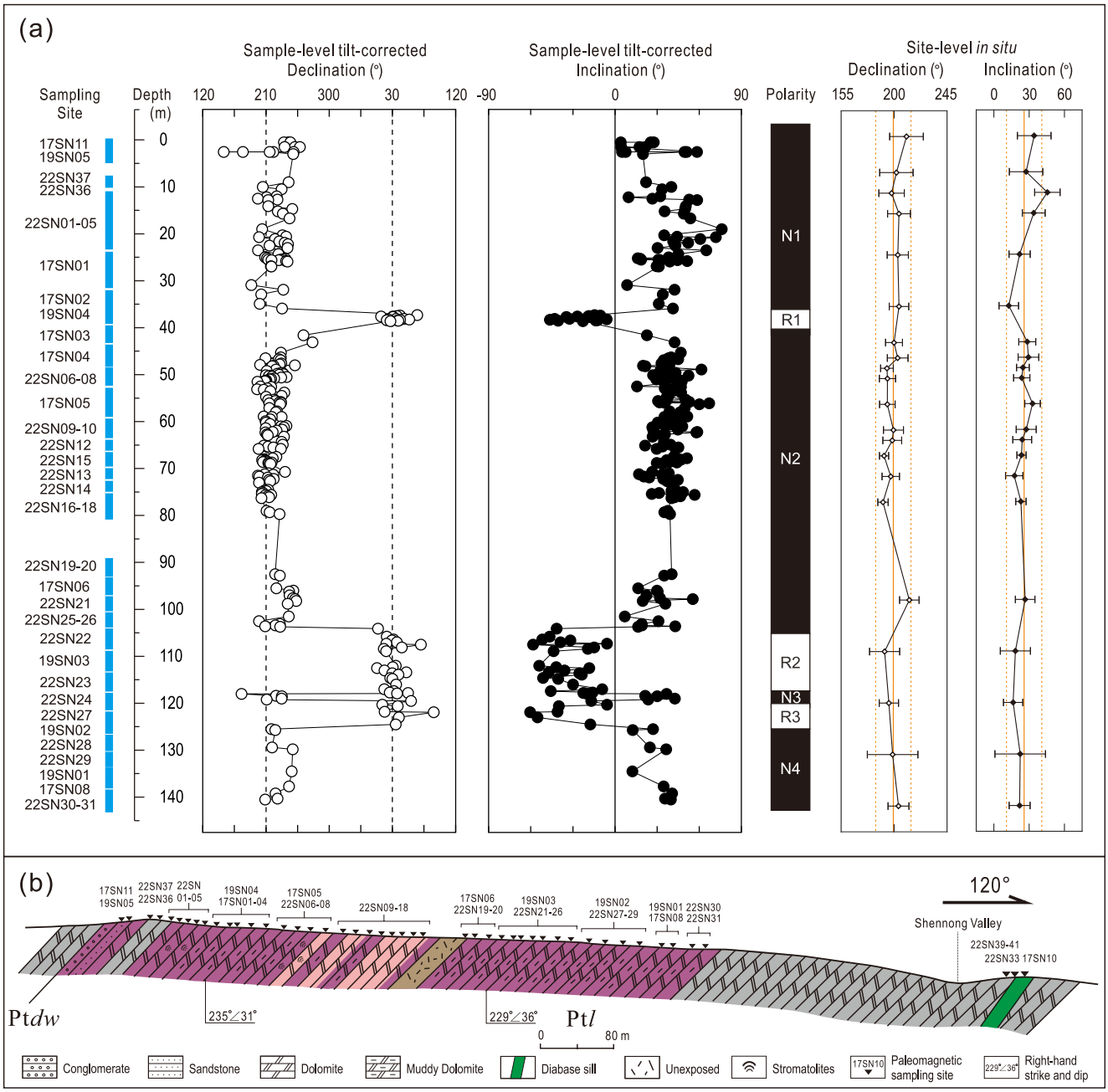


Figure 4. (a) Stratigraphic distribution of the paleomagnetic samples collected from the Luanshigou Formation, stratigraphic variation of the paleomagnetic declinations/inclinations, site-level declinations/inclinations before tilt-correction, and corresponding geomagnetic polarity zones, the character N and R represent polarity 1 and polarity 2, respectively. The mean declination/inclination of 21 sites was plotted as an orange solid line, and the ±15° interval was represented by dotted lines. (b) Sampling section shows the distributions of paleomagnetic sampling sites. Pt*dw* = Dawokeng Fm; Pt*l* = Luanshigou Fm.

between the results using the two orientation methods was less than 3° (most less than 2°), indicating no significant local magnetic anomalies in the sampling area.

3. Methods

All the paleomagnetic core samples were cut into one to three ~2.2-cm-long specimens. Rock magnetic experiments were performed on the representative parallel specimens to identify the magnetic carriers. Isothermal remanent magnetization (IRM) and back-field demagnetization of saturation IRM were conducted. IRMs were

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imparted by an ASC IM10-30 impulse magnetizer and measured using a JR-6A spinner magnetometer. Stepwise thermal demagnetization of the three orthogonal axis IRM (Lowrie, 1990) was carried out to constrain the unblocking temperatures of the magnetic carriers. Fields of 2.40 T, 0.40 T, and 0.12 T were applied to the specimens using the ASC IM10-30 pulse magnetizer along the *z*, *y*, *x* axis respectively, followed by stepwise thermal demagnetizations up to 690°C.

A total of 447 specimens, one from each sample, were subjected to stepwise thermal demagnetization experiment, using 40–50°C intervals for the steps up to 500°C, 15–30°C intervals for the steps between 500°C and 645°C, 5–10°C intervals for the steps between 645°C and 685°C, and 1–3°C intervals for the steps around 685°C. Remanent magnetizations were measured utilizing a 2G-755-4K superconducting magnetometer. Thermal demagnetization was conducted using a TD-48 furnace that has an internal residual magnetic field of less than 10 nT. Remanent magnetization measurements and demagnetization processes were carried out in a silicon steel magnetically shielded room installed at the Paleomagnetism and Environmental Magnetism Laboratory of the China University of Geosciences, Beijing. The residual field of the locations for placing the specimens in the shielded room is less than 200 nT. Magnetic components were determined through use of principal component analysis (Kirschvink, 1980). In most cases five successive steps were used for fitting a vector, with an angle deviation less than 15°. Mean directions and poles were calculated using Fisher statistics (Fisher, 1953).

4. Results

4.1. Detrital Zircon U-Pb Dating Results

A total of 170 zircon grains were managed to conduct U-Pb dating analyses. The selected zircons are mainly subrounded and rounded shape with sizes ranging from 50 to 150 μm. The cathodoluminescence images of most zircons reveal clear oscillatory growth zoning (Figure S2a in Supporting Information S1), together with the high Th/U ratios (0.4–3; Table S3), indicating their igneous origin. 161 grains with concordance over 90% yield five populations of $^{207}Pb/^{206}Pb$ ages (Figure S2c in Supporting Information S1): 2,911 ± 28 Ma (n = 10), 2674.8 ± 7.9 Ma (n = 25), 2,478 ± 20 Ma (n = 19), 2,005 ± 15 Ma (n = 23), and 1,589.5 ± 8.9 Ma (n = 67).

4.2. Rock Magnetic Results of the Luanshigou Formation

The IRM acquisition curves of the purplish-red pelitic dolomite and sandstone show a slow increase below 150 mT and display a quicker increase above 150 mT, but do not reach saturation value until 2.4 T. Progressive back-field demagnetization of saturation IRM (SIRM) reveals that the remanence coercive is over 500 mT (Figures 5d–5f), indicating the presence of high-coercivity minerals. The IRM component analysis (Kruiver et al., 2001) shows two distinct components: a low-coercivity component 1 with $B_{1/2}$ of 45–55 mT constituting less than 10% of the saturation IRM, and a high-coercivity component 2 with $B_{1/2}$ of around 540 mT constituting over 90% (Figures 5g–5i), indicating that high-coercivity minerals are predominant. The stepwise thermal demagnetization of the three orthogonal axis IRM demonstrates domination of medium and high-coercivity components with the unblocking temperature of 685°C, and the minor low-coercivity component with the unblocking temperature range of 550–580°C (Figures 5a–5c). The rock magnetic results indicate that hematite is the dominant magnetic carrier in the purplish-red pelitic dolomite and sandstone.

The IRM acquisition curves of the pale-pink dolomite specimens show a rapid increase below 100 mT and increase slower above 100 mT, but do not reach saturation value until 2.4 T. Progressive back-field demagnetization of SIRM shows that the remanence coercive is less than 60 mT (Figures 6d–6f). The data together indicate that the low-coercivity magnetic carriers are dominant, with minor high-coercivity particles existing. The IRM component analysis reveals two components: a low-coercivity component 1 with $B_{1/2}$ of around 40 mT constituting over 60% of the saturation IRM, and a high-coercivity component 2 with $B_{1/2}$ over 400 mT constituting less than 40% (Figures 6g–6i). The stepwise thermal demagnetization of the three orthogonal axis IRM demonstrates domination of the low-coercivity component with the unblocking temperature of 580°C, while the minor medium and high-coercivity components have unblocking temperatures ranging from 680 to 690°C (Figures 6a–6c). These results indicate that magnetite is the dominant magnetic carrier in the pale-pink dolomite, with the presence of hematite.

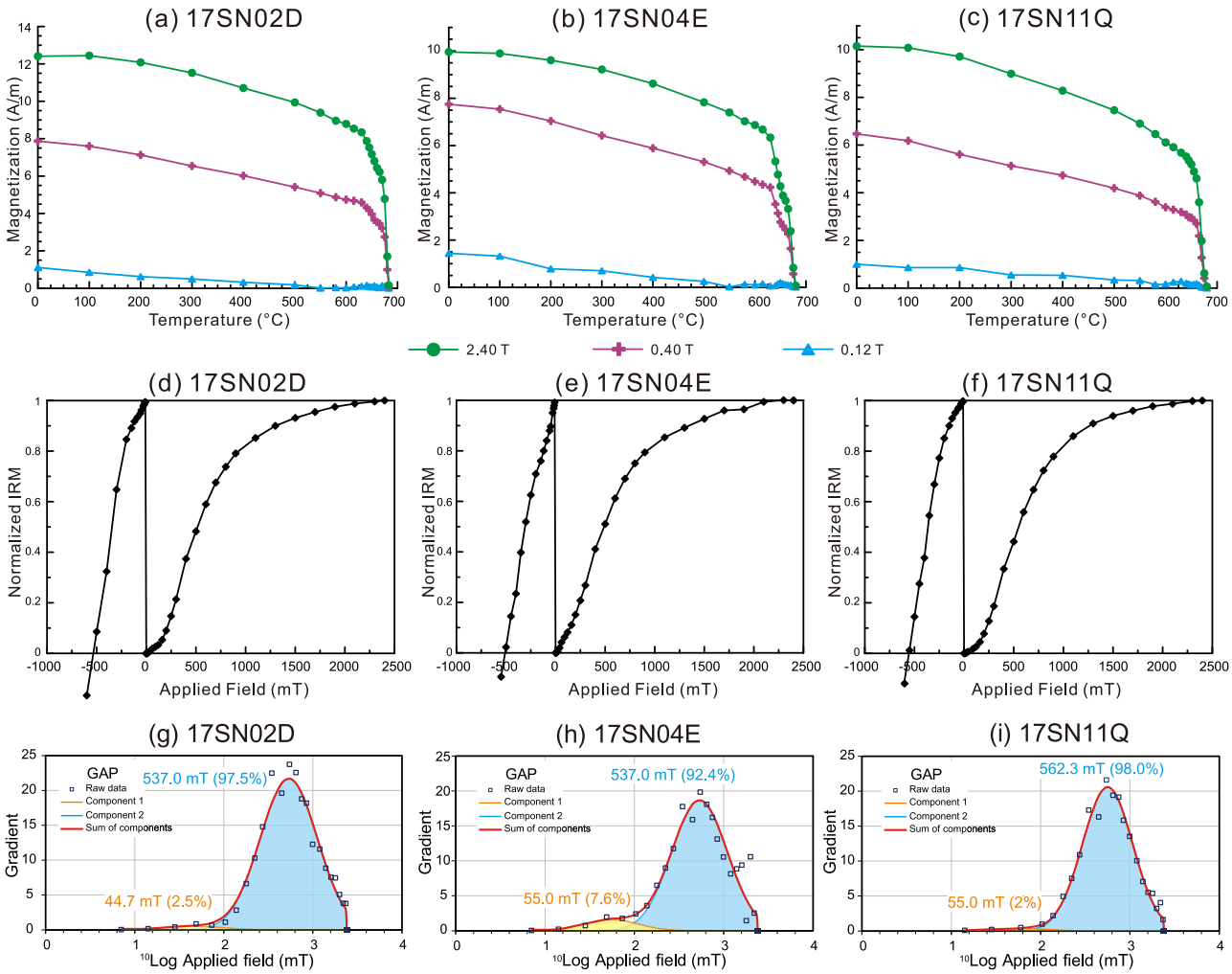


Figure 5. (a–c) Stepwise thermal demagnetizations of three orthogonal axis Isothermal remanent magnetization (IRM) of representative purplish-red pelitic dolomite and sandstone (17SN11Q) specimens. (d–f) IRM acquisition curves and back-field demagnetization curves. (g–i) IRM component analysis results. GAP: gradient acquisition plot.

4.3. Paleomagnetic Results of the Luanshigou Formation

The natural remanent magnetization (NRM) intensities of the Luanshigou Fm specimens range from 0.13 to 41.5 mA/m. Two components were isolated. The low-temperature component (LTC) that can be removed below 350–450°C has been identified in most specimens (Figure 7). Its directions in situ are close to the local geomagnetic field direction and thus were considered as a viscous remanent magnetization acquired in the recent geomagnetic field (Figure 8a).

After removed the LTC, a high-temperature component (HTC) has been obtained from 241 specimens. The HTC of the purplish-red dolomite and sandstone specimens can be isolated below the unblocking temperature of 680–690°C (Figures 7a–7c and 7f–7g), while that of the pale-pink dolomite specimens can be isolated below the unblocking temperatures of 580°C or 680°C (Figures 7d and 7e). The directions of the HTC antipodally direct southwest down and northeast up, which were termed as of polarity 1 (Figures 7a–7e) and of polarity 2 (Figures 7f and 7g), respectively. Apart from the specimens unblocked at 580°C, those of both polarities with a stable HTC display a narrow and high range of unblocking temperatures of 640–690°C. Their demagnetization curves reveal a convex shape (Figure S3 in Supporting Information S1) which is consistent with the behavior of detrital remanent magnetization (Z. Jiang et al., 2015, 2017; Swanson-Hysell et al., 2019). The specimens with HTC have revealed seven polarity zones in the sampled section with an ~8 m gap that was not sampled (Figure 4a). The HTC directions from 5 adjacent polarity zones (middle and lower N2, R2, N3, R3, and N4) passed a B-class reversal test

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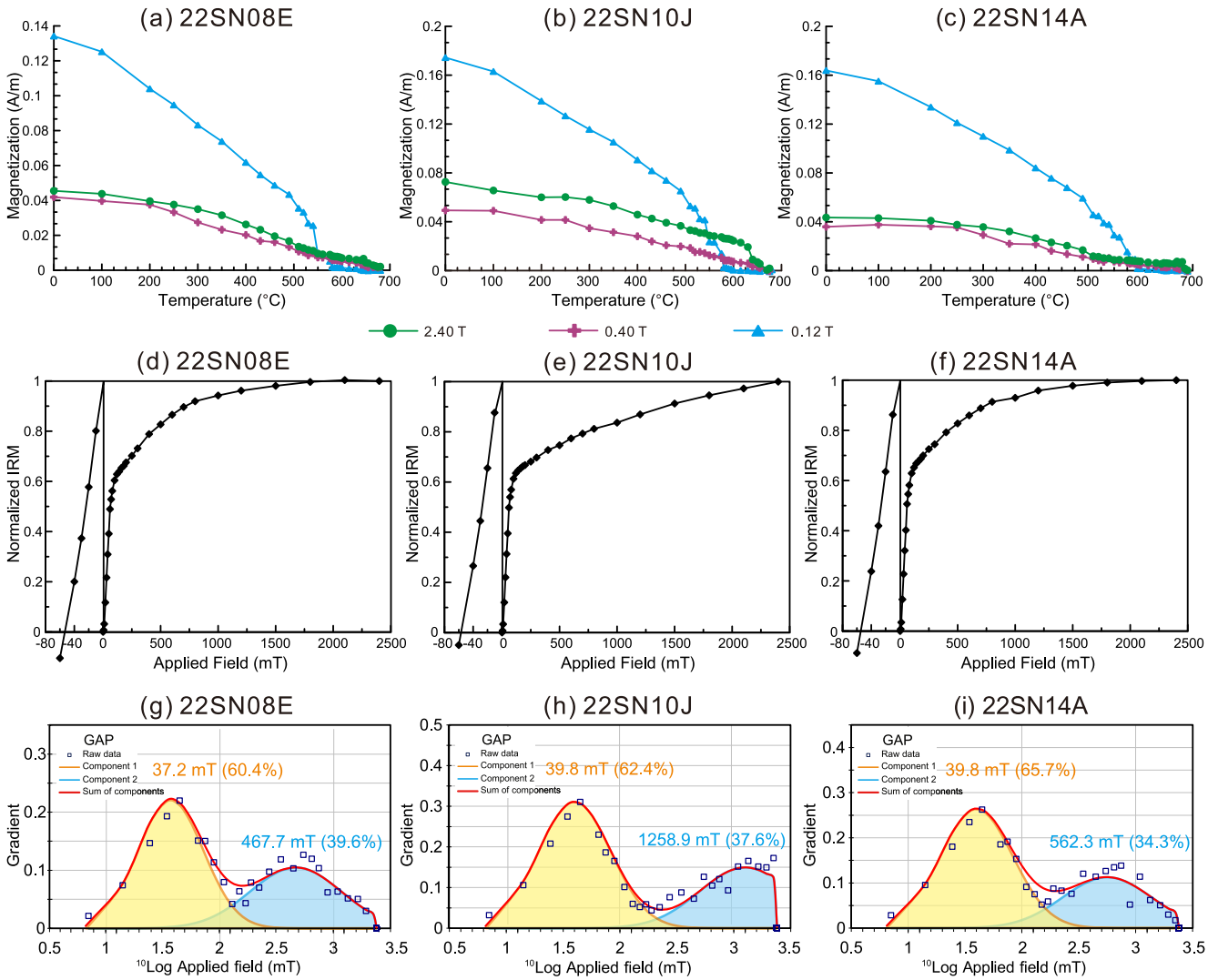


Figure 6. (a–c) Stepwise thermal demagnetizations of three orthogonal axis Isothermal remanent magnetization (IRM) of pale-pink dolomite specimens. (d–f) IRM acquisition curves and back-field demagnetization curves. (g–i) IRM component analysis results. GAP: gradient acquisition plot.

(McFadden & McElhinny, 1990; Table 1) at the 95% confidence level with an angle difference of $\gamma_o = 5.8° < \gamma_{critical} = 6.0°$. The reversal test of McFadden and McElhinny (1990) compares the means of normal and reverse directions assuming they are from the same distribution, which relies on the number of observations of each polarity and the precision parameter k. The reversal test will not pass if we include the HTC directions from the N1 and R1 polarity zones (Text S2 in Supporting Information S1), possibly due to the small sample size of R1.

We used two methods to get the mean direction of the HTC. The first method, an HTC mean direction is obtained by averaging directions of 241 individual specimens, being Dg = 199.1°, Ig = 24.8° (k = 21.1, α_{95} = 2.0°) in situ and Ds = 219.6°, Is = 37.0° (k = 21.6, α_{95} = 2.0°) after tilt correction (Table 1; Figure 8b). The second method, the HTC mean direction is obtained by averaging directions of 21 sites, being Dg = 199.3°, Ig = 25.6° (k = 81.2, α_{95} = 3.5°) in situ and Ds = 220.2°, Is = 37.3° (k = 87.7, α_{95} = 3.4°) after tilt correction (Figure S4 in Supporting Information S1). There is no significant difference between the two statistical methods (Table 1).

We used the N-dependent A_{95} envelope (Deenen et al., 2011, 2014) to determine whether the distribution of the HTC directions has sufficiently sampled the paleosecular variation (PSV). The 95% confidence envelope yielded the A_{95max} = 2.60° and A_{95min} = 1.34° for 241 specimens. The A_{95} obtained from the virtual geomagnetic poles (VGPs) of all the HTC directions is 2.0°, which falls into the A_{95} envelope. In addition, the A_{95} obtained from the

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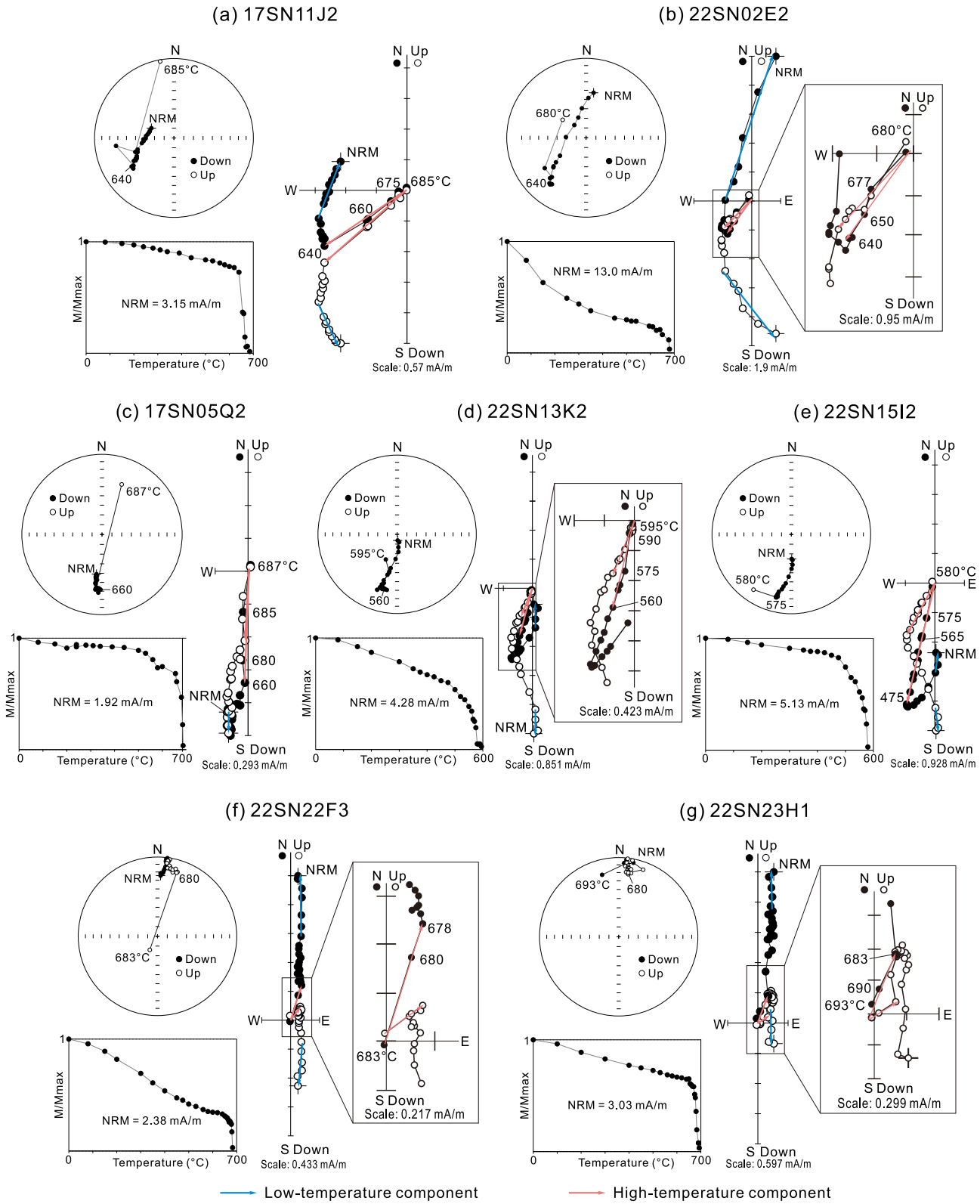


Figure 7. Orthogonal projections (Zijderveld, 1967), equal-area projections, and normalized intensity versus temperature plots of the representative specimens from the Luanshigou Formation in geographic coordinates. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. (a–e) Typical specimens of the polarity 1. (f–g) Typical specimens of the polarity 2. M = magnetization; NRM = natural remanent magnetization.

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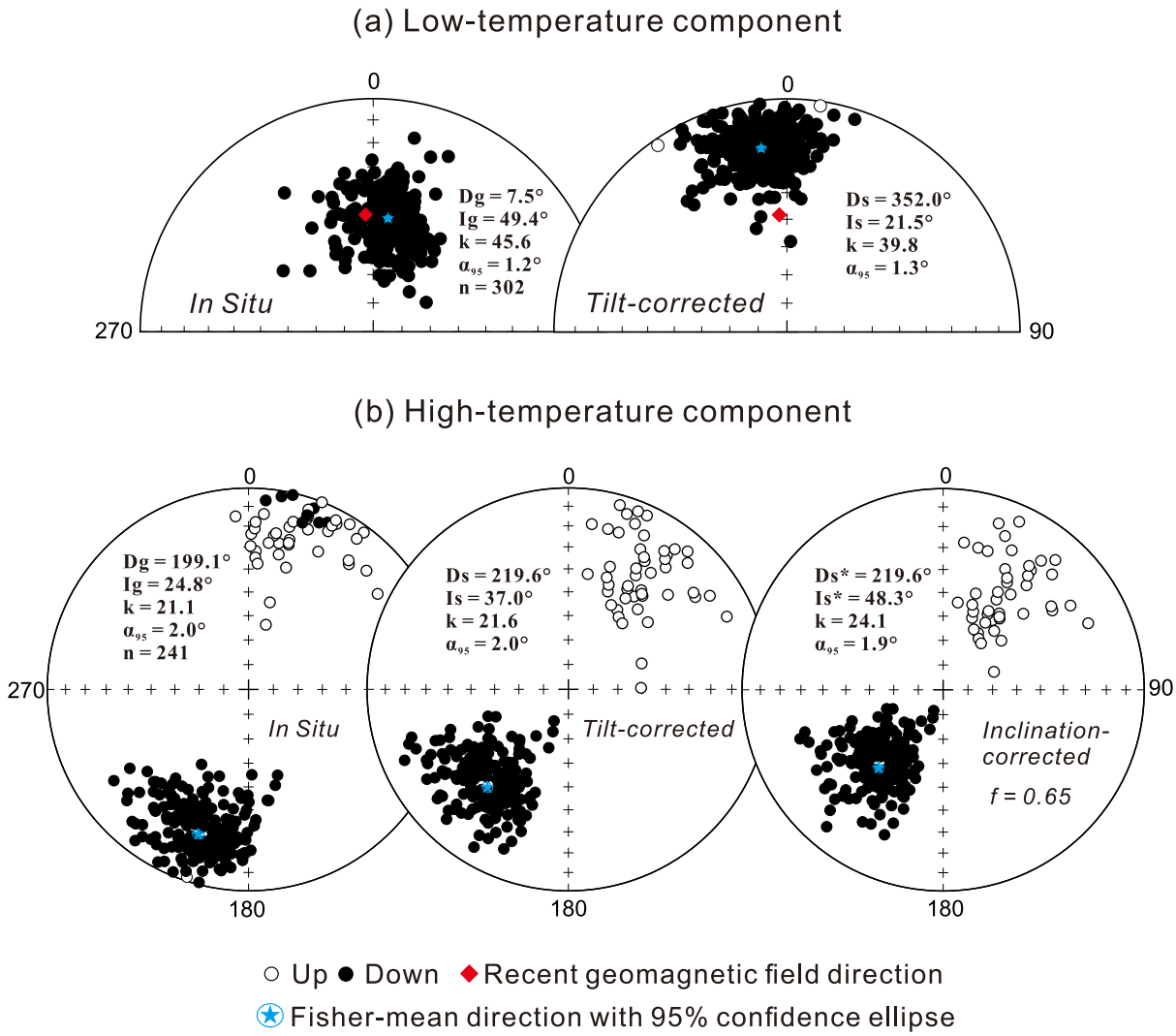


Figure 8. Equal-area projections showing paleomagnetic directions obtained by averaging directions of specimens from the Luanshigou Formation. (a) Low-temperature component. (b) High-temperature component.

sample-level VGPs of each polarity zone also fall into the corresponding A_{95} envelopes, indicating that our measured directions have sufficiently averaged out the PSV.

There are 206 specimens that revealed no stable HTC, whereas only an LTC has been isolated below 400°C. These specimens display a lower and wide range of unblocking temperatures below 670°C (Figure S3 in Supporting Information S1). The characteristics are consistent with the remagnetized hematite-bearing rocks (Z. Jiang et al., 2015, 2017). We considered these specimens unsuitable for the paleomagnetic study, and thus excluded them from further discussion.

4.4. Positive Inverse Baked Contact Test

The NRM intensities of the diabase specimens range from 1.37 to 77.2 mA/m. The directions in the low-temperature demagnetization steps were scattered, but a stable HTC has been identified between 460°C and 580°C (Figure 9b). The mean direction determined by 12 diabase specimens is Dg = 35.8°, Ig = 49.4° (k = 24.1, α_{95} = 9.4°) in situ, which is significantly different from the HTC direction of the Luanshigou Fm (Figure 9a). For the baked sandy dolomite specimens, an LTC has been isolated below 250°C. Its directions in situ are close to the local geomagnetic field direction. A HTC that directing northeast moderately down has been identified between 300°C and 450°C (Figure 9c). The mean direction obtained from the 10 baked specimens is Dg = 28.6°,

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Table 1
Paleomagnetic Results of the Luanshigou Formation in the Shennongding Section of South China

Site	n/n_0	Dg (°)	Ig (°)	Ds (°)	Is (°)	k	α_{95} (°)	Plat (°N)	Plon (°E)	dm/dp (°)	Polarity
Luanshigou Fm											
17SN11	9/23	210.5	34.3	237.9	28.8	15.9	14.3	−17.4	51.4	15.7/8.7	N
22SN05	8/13	201.9	27.4	219.3	35.3	16.2	14.2	−26.6	68.4	16.4/9.5	N
22SN02–04	9/24	197.8	45.5	230.5	52.4	23.8	10.8	−9.8	69.2	14.8/10.2	N
22SN01	12/13	204.1	33.9	226.4	41.5	20.9	9.7	−19.1	65.8	11.8/7.2	N
17SN01–02	13/13	203.1	21.9	219.2	34.4	22.3	9.0	−27.2	68.0	10.3/5.9	N
17SN02 & 19SN04	15/20	24.1	−12.7	36.0	−24.8	22.1	8.3	33.7	246.8	8.9/4.8	R
17SN03–04	8/14	199.9	28.4	224.0	39.5	59.9	7.2	−21.6	66.6	8.6/5.2	N
22SN06	10/15	203.0	29.4	227.5	38.4	31.0	8.8	−20.1	63.4	10.4/6.2	N
22SN07	14/14	193.9	24.7	215.7	40.2	55.7	5.4	−25.8	73.6	6.5/3.9	N
22SN08	12/13	194.3	23.7	215.2	39.2	40.6	6.9	−26.6	73.6	8.2/4.9	N
17SN05	12/13	194.2	32.9	223.3	46.1	44.8	6.6	−18.1	70.5	8.4/5.4	N
22SN09	10/12	199.4	27.5	222.9	39.1	33.0	8.5	−22.5	67.2	10.1/6.1	N
22SN10	11/11	198.4	24.1	219.1	37.1	33.2	8.0	−25.8	69.3	9.4/5.5	N
22SN15	13/14	191.4	23.5	212.3	40.7	115.4	3.9	−27.2	76.8	4.7/2.9	N
22SN13	11/13	197.1	17.3	213.0	32.5	38.8	7.4	−31.6	72.7	8.4/4.7	N
22SN14	14/25	190.5	23.0	211.0	40.8	81.4	4.4	−27.7	78.0	5.3/3.2	N
17SN06	15/24	212.9	26.6	233.2	30.5	22.1	8.3	−20.0	55.4	9.2/5.1	N
22SN22	10/22	11.9	−18.3	28.8	−36.2	15.1	12.8	31.5	258.2	14.9/8.7	R
22SN23	15/24	15.5	−16.4	30.8	−32.6	22.6	8.2	32.6	254.9	9.3/5.2	R
22SN27	7/16	18.7	−22.4	38.1	−35.6	8.9	21.5	27.1	249.5	24.9/14.4	R
22SN24 & 28	13/23	203.7	21.9	222.1	32.4	22.5	8.9	−26.5	64.7	10.0/5.7	N
Mean of all samples (n = 241)		***199.1***	***24.8***	***219.6***	***37.0***	***21.6***	***2.0***	***−25.5***	***68.9***	***2.3/1.4***	
Mean of all VGPs (n = 241)								***−23.9***	***68.7***	***2.0 (A_{95})***	
Mean of all sites (s = 21)		***199.3***	***25.6***	***220.2***	***37.3***	***87.7***	***3.4***	***−25.0***	***68.5***	***4.0/2.3***	
After inclination correction (f = 0.65)				***219.6***	***48.3***	***24.1***	***1.9***	***−18.5***	***74.4***	***2.5/1.6***	
Mean of all inclination-corrected VGPs (n = 241)								***−17.4***	***75.0***	***2.0 (A_{95})***	
Diabase sill											
17SN10	12/23	35.8	49.4	25.3	15.4	22.5	9.4	56.6	239.9	9.7/5.0	
Baked dolomite											
22SN39–41	10/43	28.6	57.7	22.3	21.4	84.4	5.3	61.0	240.0	5.6/2.9	

Note. n (s) = number of specimens (sites) for statistic; n_0 = total number of demagnetized specimens; Dg/Ig (Ds/Is) = declination/inclination in geographic (stratigraphic) coordinates; k = Fisher precision parameter of the mean direction; α_{95} = radius of 95% confidence circle for the mean direction; dm/dp = semi-axes of elliptical error around the pole at a probability of 95%; A_{95} = radius of 95% confidence cone of the paleomagnetic pole; Plat/Plon = latitude/longitude of the paleomagnetic pole; N/R = assigned normal/reversed polarity; VGP = virtual geomagnetic pole. Polarity zones middle and lower N2, R2, N3, R3, and N4 together passed a reversal test (McFadden & McElhinny, 1990): angle between the two antipodal mean directions $\gamma_o = 5.8° <$ critical angle $\gamma_c = 6.0°$, class B. The bold-italic values signify the mean directions of the high-temperature component and the corresponding paleomagnetic poles.

Ig = 57.7° (k = 84.4, α_{95} = 5.3°) in situ, which is close to the HTC direction of the diabase sill (Figure 9a). The results demonstrate that the host dolomite adjacent to the diabase sill was remagnetized during the emplacement of the sill, however, the strata far from the sill was not affected. We interpret that as a positive baked contact test for the sill and an inverse baked contact test for the Luanshigou Fm.

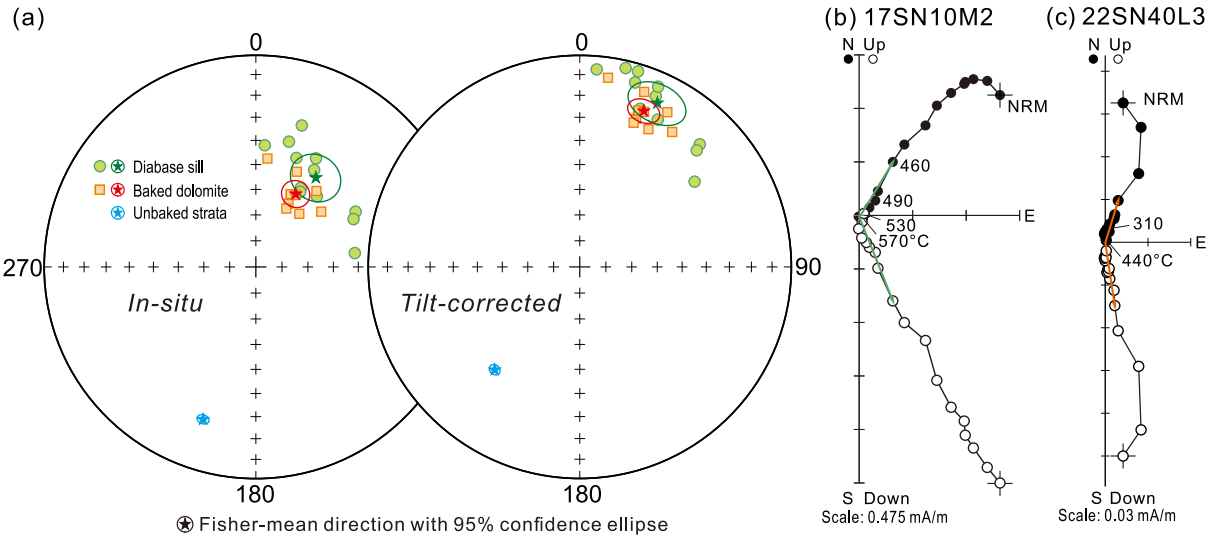


Figure 9. (a) Equal-area projections of the high-temperature component obtained from the diabase sill and the baked sandy dolomite in the Shennongding section. (b, c) Orthogonal projection (Zijderveld, 1967) in the geographic coordinate of representative diabase sill and baked specimens, respectively. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. NRM = natural remanent magnetization.

As the HTC of the Luanshigou Fm passed both a reversal test and an inverse baked contact test, we thus consider it to be a primary magnetization. We provide two options to calculate the average paleomagnetic pole of the Luanshigou Fm. Option 1, the pole was calculated from averaging the HTC directions of all specimens, being at 25.5°S, 68.9°E (dm/dp = 2.3/1.4°); option 2, the pole was obtained by averaging all the VGPs of each sample, being at 23.9°S, 68.7°E (A_{95} = 2.0°). There is no significant difference between the two options.

5. Discussion

5.1. Inclination Shallowing Correction

Paleomagnetic directions of hematite-bearing and fine-grained detrital sedimentary rocks may have suffered from depositional and/or compaction-induced inclination shallowing (Kodama, 2012; Tauxe & Kent, 2004). The relationship between the inclination observed from specimens (I_o) and the applied magnetic field inclination (I_f) can be described as the formula: tan (I_o) = f tan (I_f), where f is the flattening factor (King, 1955). In this study, the HTC directions display an east-west elongated distribution (Figure 8b), which is an important indicator of inclination shallowing (Tauxe & Kent, 2004). The elongation/inclination (E/I) method is based on the geocentric axial dipole model (GAD) developed by the data of last 5 Ma (Tauxe & Kent, 2004). Numerous studies may suggest that the geomagnetic field was GAD-dominated in most geological time since at least late Archean (e.g., Evans, 2006; Gong et al., 2023), the E/I method has been widely applied to the Precambrian paleomagnetic research (e.g., Swanson-Hysell et al., 2015; Tauxe & Kodama, 2009). We conducted the test on sample-level through use of the E/I method (Tauxe & Kent, 2004) for the HTC directions of the Luanshigou Fm. The flattening factor was calculated using the open-source software package PmagPy (https://pypi.org/project/pmagpy/). The E/I correction yielded a flattening factor of 0.65 (Figure 10), which is close to the empirical f = 0.6 for hematite-bearing sedimentary rocks (Tauxe & Kent, 1984). The inclination was corrected from 37.0° to 49.0° with a 95% confidence interval of 43.5–53.8°. The average direction after E/I correction is Ds* = 219.6°, Is* = 48.3° (k = 24.1, α_{95} = 1.9°).

The E/I correction has met all the three criteria proposed by Vaes et al. (2021): (a) we conducted the E/I correction using more than 100 individual paleomagnetic directions (we used 241 directions); (b) A_{95} = 2.0° falls within the A_{95} envelope range proposed by Deenen et al. (2011, 2014), with A_{95max} = 2.60° > 2.0° > A_{95min} = 1.34°; (c) there is no vertical-axis rotation of >15° within the data set (Figure 4a). We suggest the inclination shallowing-corrected pole as a reliable pole.

The inclination shallowing-corrected pole was calculated from the average direction and by averaging all the VGPs of each sample, being at 18.5°S, 74.4°E (dm/dp = 2.5/1.6°) and 17.4°S, 75.0°E (A_{95} = 2.0°), respectively.

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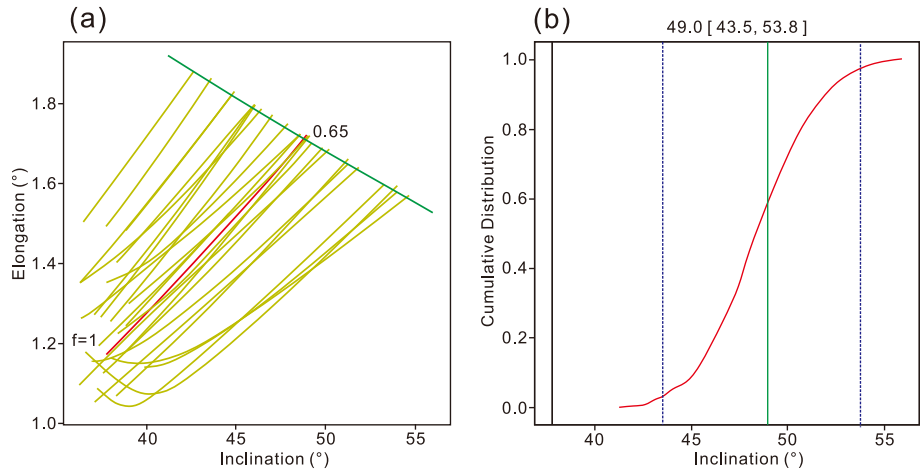


Figure 10. Elongation/inclination (E/I) correction (Tauxe & Kent, 2004) for the high-temperature component of the Luanshigou Fm. (a) Elongation/inclination as a function of flatting factor ($f = 0.65$). (b) Cumulative distribution of corrected inclination.

5.2. Age Constraints of the Luanshigou Formation

There is no age directly obtained from the Luanshigou Fm. But its age can be constrained between <1,325 Ma and 1,258.3 ± 8.5 Ma based on the detrital zircon ages of the Dayanping Fm (H. Li et al., 2013) and the Re-Os age of the Taizi Fm (B. Gao et al., 2024). The strata between the Dayanping Fm and the Taizi Fm are predominantly characterized by dolomite and have no obvious depositional hiatus. In consideration of the lithostratigraphic consistency, a minimum average sedimentary rate can be calculated by using the overall stratum thickness (~2,800 m) and a maximum time interval (~67 Ma) for the strata, which is ~4.2 cm/kyr, falling in the range of the carbonate sedimentation rates (Fairchild et al., 2016). The maximum time interval from the bottom of Taizi Fm to the Luanshigou Fm can be estimated to be ~15.9 myr. The sampled Luanshigou Fm can thus be further estimated between ~1,275 Ma and ~1,258 Ma.

The age estimation can be supported by the chemical stratigraphic data available. We compiled all the $\delta^{13}C_{carb}$ data available of the Shennongjia Gp (D. Li et al., 2022; Tian et al., 2018; Zou et al., 2019) and that from other continents. In the upper marble of the Grenville Supergroup, Laurentia, the $\delta^{13}C_{carb}$ curve displays a significant negative shift from +5 to 0%, followed by a positive shift to +2% from the bottom to the top (Whelan et al., 1990). This trend can be correlated well to the data of the Luanshigou Fm (Figure 11). The upper marble of the Grenville Supergroup has considered being younger than ~1,278 Ma based on detrital zircon age analyses (Chiarenzelli et al., 2017). Additionally, the $\delta^{13}C_{carb}$ values of the Dismal Lakes Gp range from 0% to +2%, with an increasing tendency observed in the upper section (Figure 11; Frank et al., 2003). This pattern can be compared with that of the Yingwodong and lower Dayanping formations. The Dismal Lakes Gp has been considered to be older than 1,267 ± 2 Ma (LeCheminant & Heaman, 1989) because it had been intruded by the ~1,267 Ma Mackenzie dyke swarm. The Avzyan Fm from Siberia displays $\delta^{13}C_{carb}$ values of around 0% at the bottom, which gradually increase to +4‰ in the middle section. Subsequently, there is a negative shift from +4% back to 0%, followed by a positive shift to +2% (Figure 11; Bartley et al., 2007). The Avzyan Fm is considered younger than 1,348.6 ± 3.2 Ma (Puchkov et al., 2017). It displays an overall similarity in $\delta^{13}C_{carb}$ variation with that of the Lower Shennongjia Gp. The $\delta^{13}C_{carb}$ datum correlations can in first order support the geochronological frame assigned for the Shennongjia Gp as plotted in Figure 11.

5.3. Connection Between the NYB and SW Laurentia in Late Mesoproterozoic

The paleomagnetic pole obtained from the Luanshigou Fm is distinct from any younger poles thus far obtained from south China (Figure S5 in Supporting Information S1, see S. H. Zhang et al., 2015, 2021). The data reliability is supported by an inverse baked contact test and a B-class reversal test. The pole meets nearly all the criteria of the revised seven-point criterion system (Meert et al., 2020) but a precise age of the sampled Luanshigou Fm, which can be constrained between ~1,325 Ma (H. Li et al., 2013) and 1,258.3 ± 8.5 Ma (B. Gao

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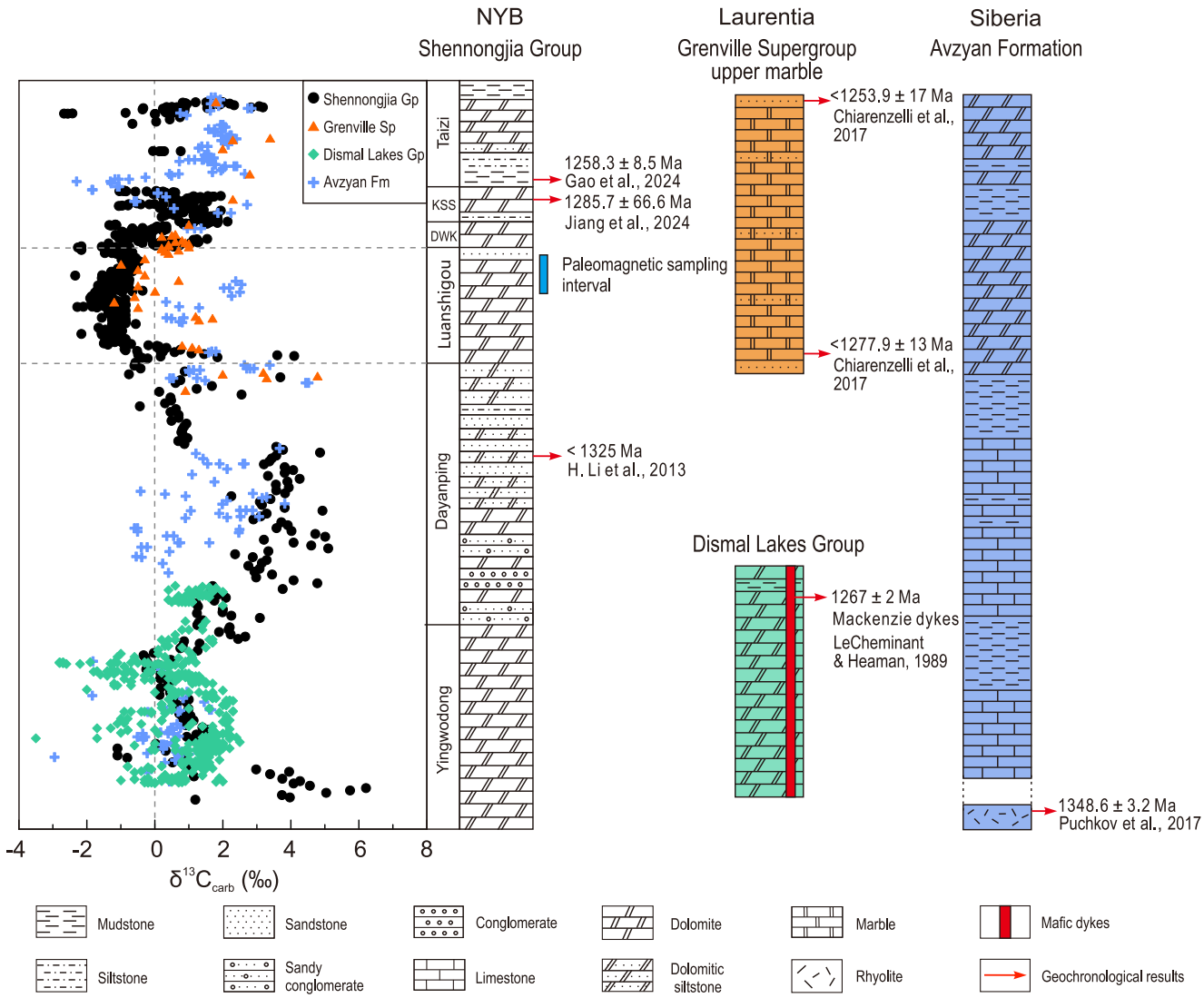


Figure 11. $\delta^{13}C_{carb}$ correlations of the Lower Shennongjia Group (D. Li et al., 2022; Tian et al., 2018; Zou et al., 2019) and coeval strata from Laurentia and Siberia (Bartley et al., 2007; Frank et al., 2003; Whelan et al., 1990). Abbreviations: NYB = northern Yangtze subblock; DWK = Dawokeng Formation; KSS = Kuangshishan Formation; Fm = Formation; Gp = Group; Sp = Supergroup.

et al., 2024), probably being at 1,270 Ma, based on the average sedimentation rate calculation and the $\delta^{13}C_{carb}$ correlation (Chiarenzelli et al., 2017; D. Li et al., 2022; Whelan et al., 1990; Zou et al., 2019). We suggest that it is a high-quality pole for the NYB.

The paleomagnetic pole of the Luanshigou Fm indicates the paleolatitude of sampling region was ~29.3°, either in southern hemisphere or northern hemisphere, given the paleomagnetic polarity uncertainty in the late Mesoproterozoic. Integrating these paleomagnetic data with regional geological evidence, we proposed a reconstruction model in which the NYB was juxtaposed to the southwestern (SW) Laurentia at the late Mesoproterozoic (Figure 12). In the reconstruction, the NYB is positioned by superimposing the Luanshigou pole atop the ~1,267 Ma pole of Laurentia (pole of Mackenzie dykes, Table 2). In fact, the apparent polar wander path of Laurentia demonstrates that there was not much polar wander in the interval between 1,320 Ma and 1,232 Ma (Figures 12a and 12c). It means the reconstruction may be valid in first order despite the Luanshigou pole may have a 35-myr age uncertainty.

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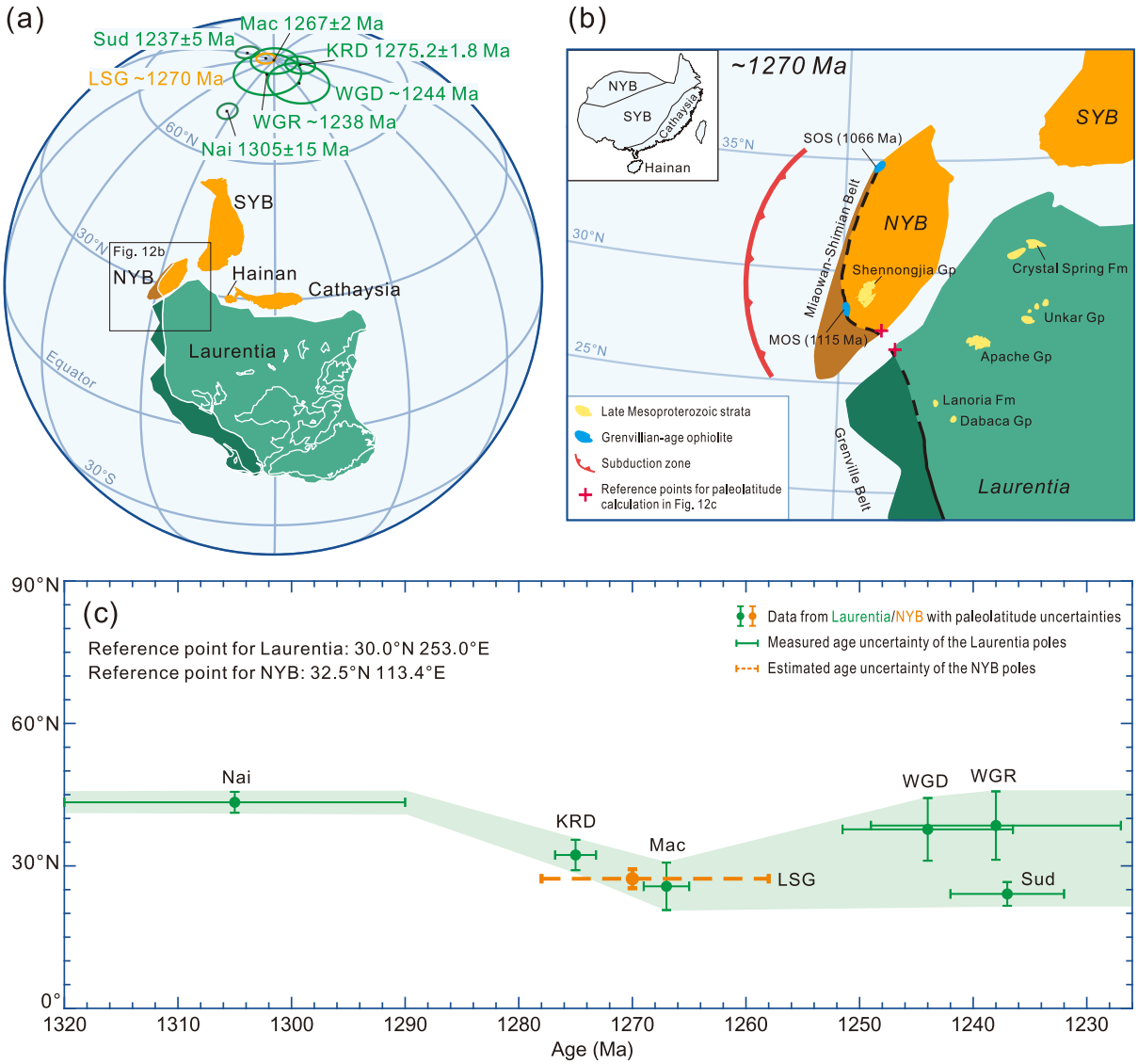


Figure 12. (a) Paleogeographic reconstructions for the northern Yangtze subblock (NYB), southern Yangtze subblock, Cathaysia, and Laurentia at ~1,270 Ma. The Paleomagnetic poles are plotted in color matching their continents and listed in Table 2. (b) Zoom in of the NYB-SW Laurentia connection with major tectonic lines (for more details see in text), the inset figure showing the present position of the tectonic subdivisions in south China. (c) Paleolatitudinal changes of Laurentia and the NYB derived from the paleomagnetic poles listed in Table 2. The age uncertainty of the Luanshigou pole was constrained between ~1,278 Ma and ~1,258 Ma based on the $\delta^{13}C_{carb}$ data correlation and the Re-Os age obtained from the Taizi Fm (for more details see in text). Abbreviations: Fm = Formation; Gp = Group; MOS = Miaowan ophiolite suite; NYB = northern Yangtze subblock; SOS = Shimian ophiolite suite; SYB = southern Yangtze subblock.

The NYB's juxtaposition with the southwestern Laurentia can be supported by the similarities of the strata developed in the two continents (Figure 13). Late Mesoproterozoic sedimentary sequences exposed in the Yavapai-Mazatzal province of SW Laurentia range in ages from 1,340 to 1,035 Ma, and recorded extensive sedimentation before and during the Grenville orogeny (Beraldi-Campesi et al., 2014; Mulder et al., 2017; Timmons et al., 2005), which are represented by Crystal Spring Fm, Unkar Gp, Apache Gp, and sedimentary sequences in New Mexico and western Texas. These Mesoproterozoic strata contain fine-grained clastic rocks and stromatolite carbonates deposited around 1,250 Ma, mafic volcanic rocks and tuffs, and were extensively intruded by crosscutting ~1,100 Ma mafic dykes (Figure 13; Amarante, 2001; Bright et al., 2014; Heaman & Grotzinger, 1992; Mulder et al., 2017). These stratigraphic and magmatic characteristics are well correlative with that of the Shennongjia region.

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Table 2
Paleomagnetic Poles Plotted in Figure 12

ID	Rock unit	Age (Ma)	Age references	Plat (°N)	Plon (°E)	A_{95} (dm/dp) (°)	Key	Paleomagnetic references
Laurentia								
Nai	Nain anorthosite	1,305 ± 15	Ryan et al. (1991)	11.7	206.7	2.2	B	Murthy (1978)
KRD	Kungnat Ring dyke	1,275.2 ± 1.8	Upton et al. (2003)	3.4	198.7	3.2	B	Piper and Stearn (1977)
Mac	Mackenzie dykes	1,267 ± 2	LeCheminant and Heaman (1989)	4.0	190.0	5.0	A	Buchan and Halls (1990)
WGD	West Gardar Dolerite dyke	1,236–1,251	Upton et al. (2003)	8.7	201.7	6.6	B	Piper and Stearn (1977)
WGR	West Gardar Lamprophyre dyke	1,227–1,249	Upton et al. (2003)	3.2	206.4	7.2	B	Piper and Stearn (1977)
Sud	Sudbury dykes	1,237 ± 5	Dudás et al. (1994)	−2.5	192.8	2.5	A	Palmer et al. (1977)
Northern Yangtze subblock								
LSG	Luanshigou Formation	~1,270	H. Li et al. (2013), B. Gao et al. (2024), D. Li et al. (2022), Zou et al. (2019)	−18.5	74.4	2.0 (2.5/1.6)		this study

Note. ID = abbreviation of the paleomagnetic pole; Plat/Plon = latitude/longitude of the paleomagnetic pole; A_{95} = radius of 95% confidence cone of the paleomagnetic pole; dm/dp = semi-axes of elliptical error around the pole at a probability of 95%; Key = Grade of the pole (Evans et al., 2021).

Detrital zircon provenance analyses suggest that the Shennongjia Gp probably shared the same provenance with the strata exposed in southwestern Laurentia, which also supports the NYB-SW Laurentia juxtaposition in late Mesoproterozoic. The detrital zircon distribution of the Shennongjia Gp displays some characteristic age peaks between 3,400 Ma and 2,000 Ma (Figure 14a), which are extensively consistent with the magmatic events identified from the basement rock units of the NYB (e.g., Kongling complex, Yudongzi complex, Houhe complex; Chen et al., 2013; S. Gao et al., 2011; Hui et al., 2017; K. Wang et al., 2018; Wu et al., 2012), indicating that the 3,400–2,000 Ma zircons came from the NYB basement. Apart from the NYB-sourced zircons, the Shennongjia Gp contains significant ~1,800–1,400 Ma detrital zircon grains (Figure 14a) that have no source area in the NYB. The 1,800–1,600 Ma Yavapai-Mazatzal province and 1,550–1,350 Ma Granite-Rhyolite province in southwestern Laurentia (Mulder et al., 2017) have been considered as the major provenance of the Mesoproterozoic strata in the SW Laurentia (Figures 14b and 14c) might also provide the detrital zircon grains for the Shennongjia Gp.

In tectonics, the Miaowan-Shimian suture zone is likely a continuation of the Grenville belt. Magmatic zircons from a deformed gabbro in the Miaowan ophiolite suite (MOS) yield an age of 1,115 ± 29 Ma, thus the MOS was interpreted as being formed at ~1,115 Ma (Deng et al., 2017). Some subduction-related ~1,000–970 Ma gabbroic rocks and ~1,050–1,020 Ma granitic rocks were also recognized from the Miaowan melange (Deng et al., 2012, 2017; X. Jiang et al., 2016) in the southern margin of NYB. Zircon U-Pb dating on a gabbro sample of the Shimian ophiolite yielded an age of 1,066 ± 11 Ma, representing the formation age of the Shimian ophiolite (P. Hu et al., 2017; J. H. Zhao et al., 2017). The subduction ages of the Miaowan-Shimian suture zone are broadly consistent with that in the Grenville belt along the southern margin of Laurentia, which recorded ~1,360–1,080 Ma orogenic activity (Mosher, 1998; Mosher et al., 2008). We thus suggest that the Miaowan-Shimian suture zone was likely a continuation of the Grenville belt based on their similar age and the same paleolatitude derived from the paleomagnetic results.

We also noticed the model in which Cawood et al. (2020) suggested that the Yangtze block was once probably juxtaposed to the northern Laurentia in Mesoproterozoic. The ~1,267 Ma Mackenzie dyke swarms (LeCheminant & Heaman, 1989), which were exposed extensively in northwestern Laurentia, have not been found in south China, therefore, we do not favor putting south China blocks close to northern Laurentia at ~1,270 Ma.

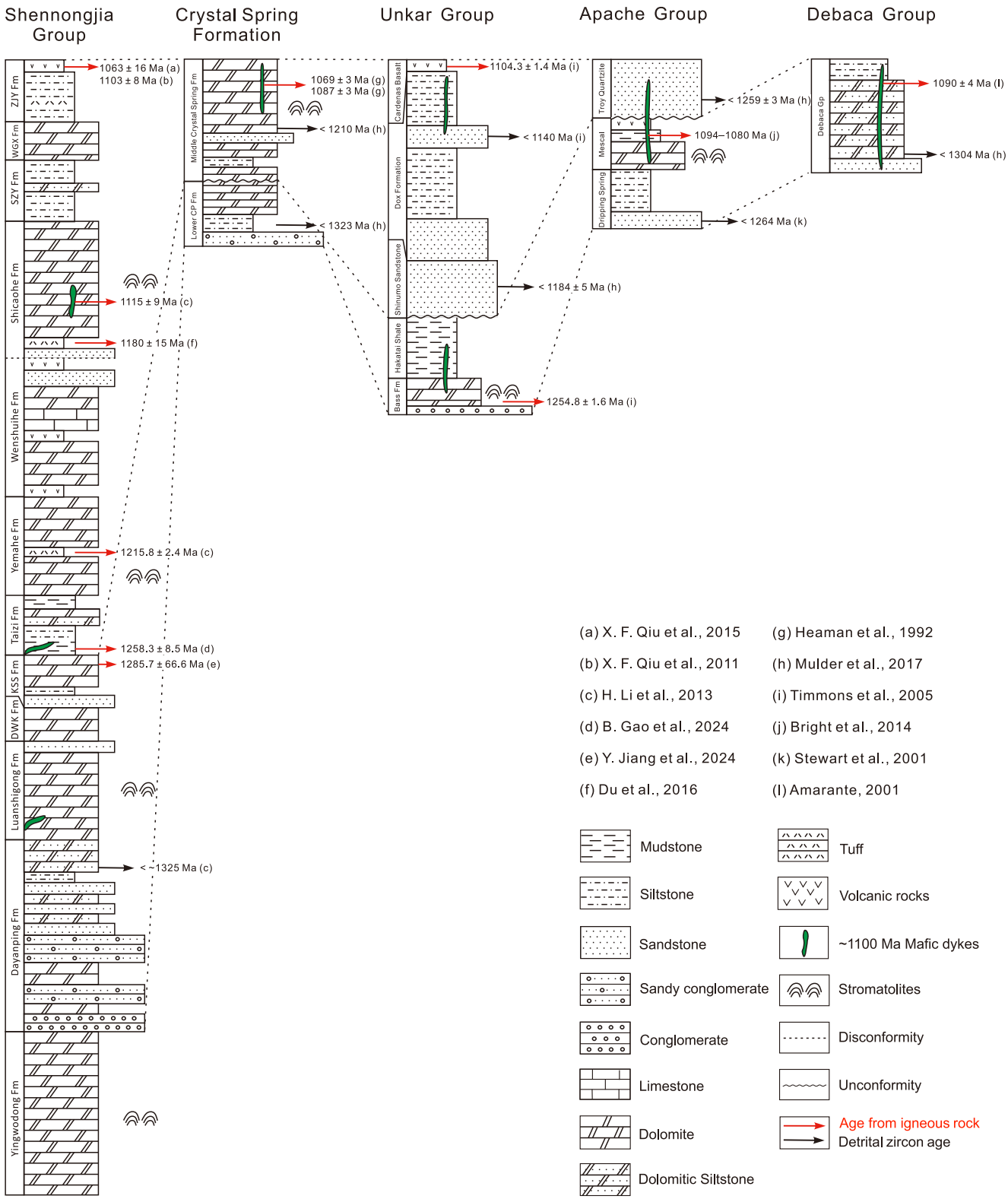


Figure 13. Stratigraphic similarities between the Shennongjia Group (in northern Yangtze subblock) and Crystal Spring Formation, Unkar Group, Apache Group, Debaca Group (in SW Laurentia), and ~1,100 Ma mafic dykes intruded in them. CP = Crystal Spring; DWK = Dawokeng; Fm = Formation; KSS = Kuangshishan; SZY = Songziyuan; WGX = Wagangxi; ZJY = Zhengjiaya.

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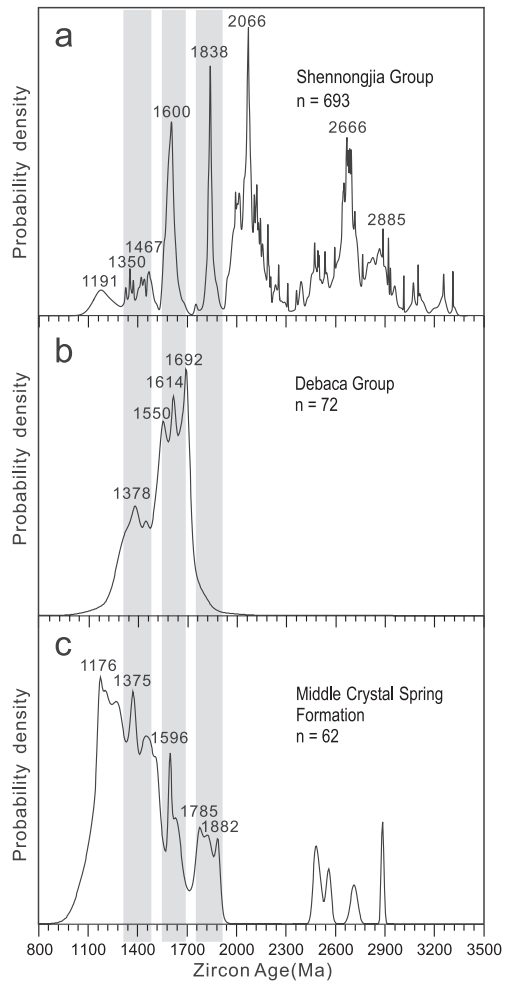


Figure 14. Probability density plots of detrital zircon age. (a) Data of the Shennongjia Group (H. Li et al., 2013; Xiao, 2012; D. Xu et al., 2016; J. Wang et al., 2013; this study). (b–c) Data of SW Laurentia (Mulder et al., 2017).

If our model is correct, then the NYB should be located in the northern hemisphere with Laurentia during the late Mesoproterozoic. For this reason, the southwestward, moderately down direction of the HTC of the Luanshigou Fm is assigned as of normal paleomagnetic polarity, and its antipodal, the northeastward, moderately up direction, as of reversed polarity.

5.4. Paleogeographic Positions of the SYB and Cathaysia

As noticed by many researchers, similar geological records may strongly suggest that the Hainan terrane, traditionally assumed to be part of Cathaysia, was probably connected with western Laurentia in the Nuna supercontinent (Z. X. Li et al., 1995, 2008; Y. J. Xu et al., 2020; S. H. Zhang et al., 2012). However, Cawood et al. (2020) argued that the Hainan terrane may have been part of the SYB, rather than the Cathaysia block, based on the provenance similarities of the Shilu Gp in the Hainan island and the Kunyang Gp in the SYB. This viewpoint may support the model of the tight neighborhood between the SYB and SW Laurentia, but raise more uncertainty of the Cathaysia's affiliation in the Nuna supercontinent.

There has been not much strong evidence to determine the SYB's paleogeographic position yet. Because the assembly of the NYB and SYB has been accomplished by ~860 Ma (Lu et al., 2020) and the NYB and SYB have similar tectonic evolution history since at least late Paleoproterozoic (Cawood et al., 2020), it is likely that the two blocks were located not far from each other in the context of the coherent Nuna in Mesoproterozoic. We thus tentatively placed the SYB also close to SW Laurentia around ~1,270 Ma in the paleogeographic reconstruction (Figure 12a).

6. Conclusions

A new late Mesoproterozoic paleomagnetic pole (18.5°S, 74.4°E, dm/dp = 2.5/1.6°) was obtained from the Luanshigou Fm of Shennongjia Gp in the NYB, with quality of this pole was assured by a positive inverse backed contact test and a B-class reversal test. The age of the pole is constrained between ~1,325 Ma and ~1,258 Ma. This new pole suggests a ~30° paleolatitudinal position for the NYB around 1,270 Ma, and supports a Pre-Grenvillian NYB-SW Laurentia connection, in which the late Mesoproterozoic Miaowan-Shimian suture zone was likely the continuation of the Grenville belt.

Data Availability Statement

The data used in this study (Liu et al., 2024) are available in Supporting Information S1, Tables S2 and S3. PaleoMac 6.5 software package (Cogné, 2003a, 2003b) was used for analyzing the paleomagnetic data. PmagPy 4.2.92 software package (Tauxe et al., 2016, 2023) was used for inclination shallowing correction.

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